

Prompt Engineering and Large Language Models

Makeup Examination

Whole course, weighted towards Weeks 6–13 · 25 questions × 4 points = 100 points

Name: _____ Student No: _____ Date: _____

Instructions. This paper contains 25 multiple-choice questions. Select the **single best** answer for each. Each question is worth equal marks. Time allowed: 60 minutes. No notes or electronic devices.

1. Which is the single operation underlying everything a large language model produces?
 - A) Matching the question against a database of answers
 - B) Predicting the next token, repeatedly
 - C) Executing rules supplied by developers
 - D) Summarising retrieved web pages
2. Why is the model unable to signal when it is guessing?
 - A) Guessing is suppressed by safety training
 - B) The same operation runs whether the pattern is strong or weak, and confidence is not part of the output
 - C) Confidence scores exist but are hidden by the interface
 - D) Users have disabled the feature
3. A model is asked about a regulation amended after its knowledge cut-off. What is most likely?
 - A) It reports that its knowledge ends at a specific date
 - B) It answers confidently using the earlier version or something adjacent
 - C) It refuses to answer
 - D) It searches for the update automatically
4. Tokenisation explains all of the following **except**:
 - A) Difficulty counting letters within a word
 - B) Inconsistent behaviour of word-count instructions
 - C) Higher cost for Turkish text than English
 - D) The tendency to accept a false premise in a question
5. What happens when a document exceeds the context window in the most dangerous case?
 - A) The service returns an explicit error
 - B) Part of the material is dropped silently and a confident answer follows
 - C) The answer is visibly cut off
 - D) The model asks for a shorter document
6. Which statement about fine-tuning for safety is accurate?
 - A) It installs a rule that blocks disallowed output

- B) It shifts probabilities, producing a tendency rather than a guarantee
 - C) It removes disallowed material from training data
 - D) It applies only to images
7. Which lever should be pulled when the output assumes the wrong reader or the wrong country?
- A) Format
 - B) Examples
 - C) Context
 - D) Temperature
8. Why does specifying a fallback string reduce invented content?
- A) It lowers the temperature
 - B) It gives the honest answer a fixed and probable form
 - C) It blocks access to training data
 - D) It shortens the response
9. A summary is factually correct throughout but omits the only finding that would have changed a decision. This is:
- A) Fabrication
 - B) Displacement
 - C) Omission
 - D) Over-reading
10. An answer about local practice accurately describes the practice of a different country. This is:
- A) Displacement
 - B) Fabrication
 - C) Silent failure
 - D) Leakage
11. An assistant grounded in your documents cites a genuine sentence that does not support the claim it is attached to. This is:
- A) Answered from memory
 - B) Retrieved the wrong passage
 - C) Omission
 - D) Fabrication
12. Your source states a requirement applies “in exceptional cases”. The assistant reports it applies generally. This is:
- A) Over-reading the passage
 - B) Fabrication
 - C) Displacement
 - D) Retrieved the wrong passage
13. An automation produces a polished result from corrupted input, with no error anywhere. This is:

- A) Wrong stopping point
 - B) Silent failure
 - C) Context overflow
 - D) Direct injection
- 14.** Which is genuine verification of a factual claim?
- A) Asking the model how confident it is
 - B) Asking the same question a second time
 - C) Locating the claim in a source you trust
 - D) Asking the model to double-check its work
- 15.** A generated tool works on the inputs you tried and fails on an empty field. Which failure is this?
- A) Wrong logic with right appearance
 - B) Correct only for the cases tried
 - C) Silently discarded input
 - D) Indirect injection
- 16.** Why is a consistently wrong generated tool more dangerous than an intermittently wrong one?
- A) It uses more resources
 - B) Consistency resembles reliability, so the error is not noticed
 - C) It is harder to regenerate
 - D) It fails at higher volumes
- 17.** Five steps at 95% reliability gives an end-to-end reliability closest to:
- A) 95%
 - B) 90%
 - C) 77%
 - D) 50%
- 18.** Why does the “am I finished?” decision cause difficulty for agents?
- A) It requires access to external tools
 - B) It is a judgment about matching an intention that is not in the context, so it is predicted like anything else
 - C) It consumes excessive tokens
 - D) It is disabled by default
- 19.** A workflow drafts patient letters and sends them automatically. Which design rule is violated?
- A) Keep the chain short
 - B) Make each step inspectable
 - C) Put the human before the irreversible action
 - D) Use the narrowest tools
- 20.** Which is the best candidate for a first AI workflow?

- A) The most intellectually demanding task in your field
 - B) A repetitive task whose answers exist in a document, where a mistake is recoverable
 - C) A task requiring access to personal records
 - D) A task nobody currently performs
- 21.** Answering “users might receive incorrect information” to the question of who is harmed is inadequate because:
- A) It is too brief
 - B) It commits to nothing and therefore changes no design decision
 - C) Users are not the only stakeholders
 - D) It omits the probability of harm
- 22.** Which does a Domain Pack **not** address?
- A) Institution-specific vocabulary
 - B) The structure professionals expect
 - C) What must never be stated without qualification
 - D) Whether the content is true
- 23.** What makes prompt injection difficult to eliminate?
- A) Attackers are highly skilled
 - B) Instructions and data arrive through the same undifferentiated channel
 - C) Context windows are too small
 - D) Safety training is incomplete
- 24.** Which attacker objective becomes possible only when tools are connected?
- A) Altering the content of an answer
 - B) Extracting other documents from the context
 - C) Sending, deleting or purchasing
 - D) Revealing system instructions
- 25.** Which statement about anonymising clinical material is correct?
- A) Removing names is sufficient
 - B) Rare details in a small population can identify a person even without names
 - C) Anonymised data may be uploaded anywhere
 - D) Anonymisation applies only to published research